



Joint Model Importance via the Rashomon Set

Behavioural Comparison of Near-Equally-Good Models

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May 29, 2026

Beyond the single best model

- In tabular ML, many models reach indistinguishable predictive performance.
- For a tolerance $\varepsilon > 0$, the *Rashomon set*

$$\text{RS}(\varepsilon) = \{ \hat{f}_{\mathcal{D},\lambda} : \mathcal{R}(\hat{f}_{\mathcal{D},\lambda}) \leq \mathcal{R}(\hat{f}_{\mathcal{D},\lambda^*}) \cdot (1 + \varepsilon), \lambda \in \Lambda \}$$

collects every model within ε of the best.

Notation: $\hat{f}_{\mathcal{D},\lambda}$ = model trained on data $\mathcal{D}$ with hyperparameter configuration $\lambda \in \Lambda$; λ^* = risk-minimizing configuration; $\mathcal{R}(\cdot)$ = risk.

- Members of $\text{RS}(\varepsilon)$ can disagree substantially on individual predictions and on feature importance.
- **Question.** How are these equally good models related to one another, and what does the collective tell us beyond the single best?

- 1 For every $\hat{f}_{\mathcal{D},\lambda} \in \text{RS}(\varepsilon)$, compute the prediction vector $\hat{y}$ on a held-out 2/3 validation split.
- 2 Build the pairwise *behavioural distance* matrix D with entries

$$d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p = \left(\sum_{o=1}^{n_{\text{val}}} |\hat{y}_{i,o} - \hat{y}_{j,o}|^p \right)^{1/p},$$

where i, j index models in $\text{RS}(\varepsilon)$ and o runs over the n_{val} validation observations.

- 3 View D three ways:
 - ▶ hierarchical clustering (`hclust`)
 - ▶ classical MDS (`cmdscale`, embedding dimension $k = 2$)
 - ▶ partitioning around medoids (`cluster::pam`)

Distances computed for both Euclidean ($p = 2$) and Manhattan ($p = 1$); results shown are Euclidean (Manhattan tells the same story).

Rashomon tolerance $\varepsilon = 1\%$ of the minimum risk.

Five learner families considered for every task: `tree`, `glmnet`, `xgb`, `nnet`, `svm`.

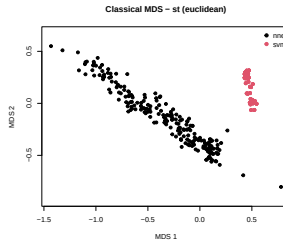
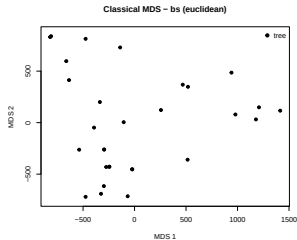
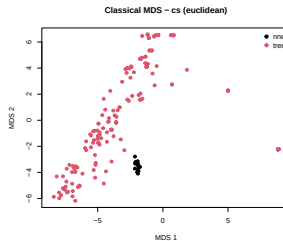
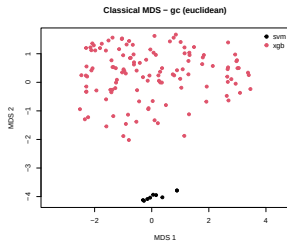
Key	Task	Type
gc	German Credit	classification
cs	COMPAS	classification
bs	Bike Sharing	regression
st	Synthetic	regression

Who survives the Rashomon cut?

Task	Surviving families (n models)
gc	svm(9), xgb(125)
cs	nnet(15), tree(261)
bs	tree(30)
st	nnet(216), svm(160)

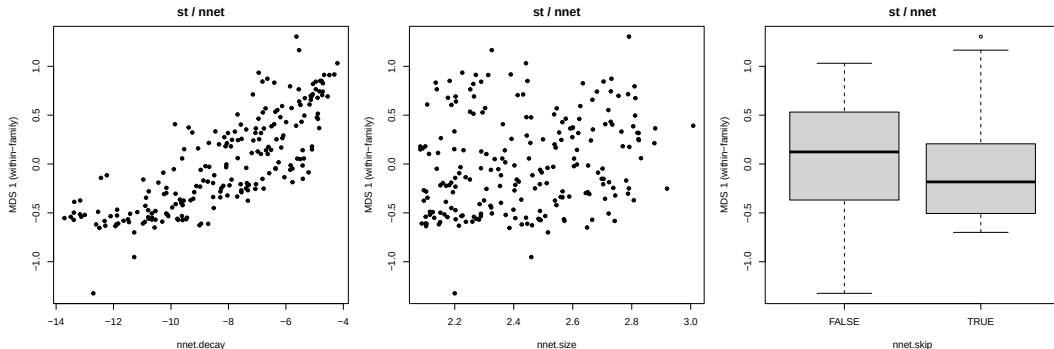
First finding. At $\varepsilon = 1\%$ only one or two learner families survive per task — *learner choice* is the first dimension of disagreement inside $RS(\varepsilon)$.

MDS view — all four tasks



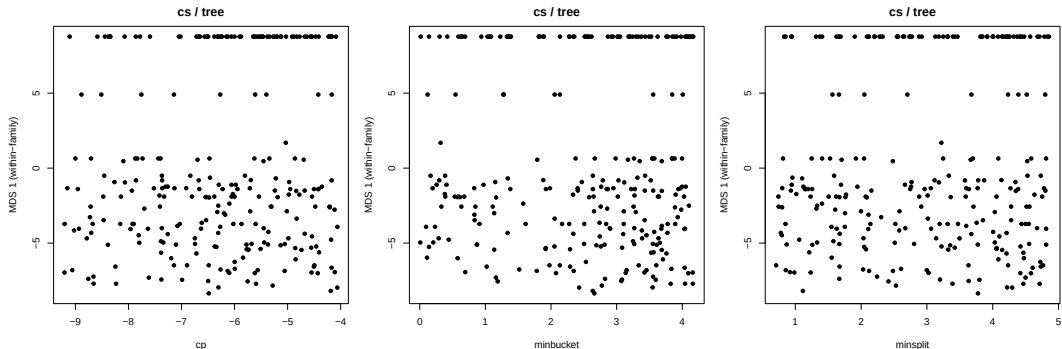
- Wherever two families coexist, behavioural distance **separates them cleanly** — the dominant axis is family identity, not within-family variation.
- Within-family variance is *asymmetric*: `svm` and `nnet` form tight clusters; `tree` and `xgb` scatter widely despite fewer tunable hyperparameters.
- Several families trace out a **1D-manifold** in the MDS plane (e.g. `st/nnet`, `cs/tree`) — next two slides.

Within-family structure I — st / nnet



- Each panel: 1D within-family MDS coordinate vs. one hyperparameter.
- The within-family manifold **is the decay axis** — near-monotone mapping.
- size and skip have no effect on behavioural position.

Within-family structure II — cs / tree



- Within-family MDS coordinate vs. cp, minbucket, minsplit.
- Trees on COMPAS are **bimodal**: a tight cluster at $\text{MDS } 1 \approx 8$ and a diffuse population below zero.
- *No single tuned hyperparameter* separates the two modes. **Speculation**: a discrete behavioural split driven by structure (e.g. root split, depth) that hyperparameter tuning does not reach — to be confirmed.

Scope — behaviour is not importance

- **What we have computed:** a *behavioural* distance $d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p$ from validation predictions.
- **What “joint model importance” asks:** how does the per-model feature-importance vector $\phi^{(i)}$ vary across $i \in \text{RS}(\varepsilon)$? (Variable-Importance Cloud, VIC.)
- **The two are independent axes.**
 - ▶ Identical predictions ($d_{ij} = 0$) can hide different feature reliance.
 - ▶ Different predictions need not produce different importance profiles.
- Today: the behavioural skeleton of $\text{RS}(\varepsilon)$ — a prerequisite, not an answer, for joint importance.
Next: rerun the pipeline with a VIC-based distance and compare cluster structure.

- The behavioural skeleton of $\text{RS}(\varepsilon)$ has two-level structure:
 - ① Which learner family does the model come from?
 - ② Where on the family's behavioural manifold does it sit?
- Across every task observed, learner choice is the first axis of *behavioural* disagreement — not hyperparameter setting.
- Within-family behaviour can be smooth and HP-aligned (`st/nnet` $\sim$ decay) or discrete and HP-orthogonal (`cs/tree`; *causal mechanism still speculative*).
- **Open:** does *importance* follow the same structure? — VIC-based distance is the next step.

- **ε sensitivity.** How does family composition shift as the Rashomon tolerance opens?
- **Feature-importance distance.** Compare behavioural distance with a distance built from VIC — do the clusters agree?
- **Choose the number of clusters k for PAM via silhouette,** then read out the medoids as “representative” members of $RS(\varepsilon)$.
- **Disagreement map.** Where in feature space do Rashomon-set models disagree the most?
- **Discrete tree mode on cs.** Is the bimodality driven by structural differences (root split, depth) that hyperparameters do not capture?

Thank you.

Questions?

Backup — the behavioural distance matrix in detail

Stack the Rashomon-set prediction vectors row-wise into a prediction matrix:

$$\mathbf{P} = \begin{pmatrix} \hat{y}_{1,1} & \hat{y}_{1,2} & \cdots & \hat{y}_{1,n_{\text{val}}} \\ \hat{y}_{2,1} & \hat{y}_{2,2} & \cdots & \hat{y}_{2,n_{\text{val}}} \\ \vdots & \vdots & & \vdots \\ \hat{y}_{n,1} & \hat{y}_{n,2} & \cdots & \hat{y}_{n,n_{\text{val}}} \end{pmatrix} \in \mathbb{R}^{n \times n_{\text{val}}}$$

with one row per model in $\text{RS}(\varepsilon)$ (here $n = |\text{RS}(\varepsilon)|$) and one column per held-out validation observation (here n_{val}).

Row-wise L_p distances collapse the observation axis and yield the symmetric matrix

$$\mathbf{D} \in \mathbb{R}^{n \times n}, \quad d_{ij} = \|\hat{y}_i - \hat{y}_j\|_p.$$

Rows / columns of $\mathbf{D}$ now index *models*, not observations. Hence **one MDS point = one model**; the cloud sits in $\mathbb{R}^n$ and is visualised in $\mathbb{R}^2$ via `cmdscale`.